$C_{d\phi_1}^{(22)\,i1\_i2\_} \to \frac{1}{16\pi^2}\Big(\frac{1}{3} m_2 \tilde{\mu} g_2^4 \left(-3 y_d^{i1i2} + 2 y_d^{pi2}\delta_{pi1}\right) LF_{2,2,0}[m_2, \tilde{\mu}] +$
$\frac{1}{6} m_2 \tilde{\mu} g_2^4 \left(3 y_d^{i1i2} - 2 y_d^{pi2}\delta_{pi1}\right) LF_{3,2,-1}[m_2, \tilde{\mu}] -$
$\frac{1}{4} \tilde{\mu} g_1^2 y_d^{pr} y_d^{i1i2} \overline{a_d}^{pr} LF_{2,2,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] - \frac{1}{4} \tilde{\mu} g_1^2 y_d^{i1i2} y_e^{pr} \overline{a_e}^{pr} LF_{2,2,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] +$
$\frac{1}{2} \tilde{\mu} g_1^2 y_d^{i1i2} y_e^{pr} \overline{a_e}^{pr} LF_{3,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \frac{1}{4} \tilde{\mu} g_1^2 y_d^{i1i2} y_e^{pr} \overline{a_e}^{pr} LF_{3,2,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] -$
$\frac{1}{8} \tilde{\mu} y_e^{pr} \overline{a_e}^{pr} \left(y_d^{i1i2}(9 g_1^2 + g_2^2) - 2 g_2^2 y_d^{si2}\delta_{si1}\right) LF_{4,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] +$
$\frac{1}{6} \tilde{\mu} y_e^{pr} \overline{a_e}^{pr} \left(3 g_1^2 y_d^{i1i2} - 2 g_2^2 y_d^{si2}\delta_{si1}\right) LF_{5,1,-2}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] +$
$\frac{1}{2} \tilde{\mu} g_1^2 y_d^{pr} y_d^{i1i2} \overline{a_d}^{pr} LF_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] + \frac{1}{4} \tilde{\mu} g_1^2 y_d^{pr} y_d^{i1i2} \overline{a_d}^{pr} LF_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] -$
$\frac{3}{8} \tilde{\mu} y_d^{pr} \overline{a_d}^{pr} \left(y_d^{i1i2}(5 g_1^2 + g_2^2) - 2 g_2^2 y_d^{si2}\delta_{si1}\right) LF_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] +$
$\frac{1}{2} \tilde{\mu} y_d^{pr} \overline{a_d}^{pr} \left(3 g_1^2 y_d^{i1i2} - 2 g_2^2 y_d^{si2}\delta_{si1}\right) LF_{5,1,-2}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] -$
$\frac{1}{8} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} \left(y_d^{i1i2}(g_1^2 + 9 g_2^2) - 4 g_2^2 y_d^{si2}\delta_{si1}\right) LF_{2,2,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$
$\frac{3}{4} \tilde{\mu} g_2^2 y_d^{i1i2} y_u^{pr} \overline{a_u}^{pr} LF_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] - \frac{1}{4} \tilde{\mu} g_2^2 y_d^{si2} y_u^{pr} \overline{a_u}^{pr} LF_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}]\,\delta_{si1} -$
$\frac{3}{4} \tilde{\mu} g_2^2 y_d^{i1i2} y_u^{pr} \overline{a_u}^{pr} LF_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$
$\frac{1}{4} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} \left(7 g_1^2 y_d^{i1i2} - 2 g_2^2 y_d^{si2}\delta_{si1}\right) LF_{3,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] +$
$\frac{1}{8} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} \left(y_d^{i1i2}(g_1^2 + 9 g_2^2) - 4 g_2^2 y_d^{si2}\delta_{si1}\right) LF_{3,2,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] +$
$\frac{3}{4} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} \left(-5 g_1^2 y_d^{i1i2} + 2 g_2^2 y_d^{si2}\delta_{si1}\right) LF_{4,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] +$
$\frac{1}{2} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} \left(3 g_1^2 y_d^{i1i2} - 2 g_2^2 y_d^{si2}\delta_{si1}\right) LF_{5,1,-2}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] +$
$\frac{1}{4} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 - g_2^2) LF_{3,1,0}[\tilde{\mu}, m_1] + \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} LF_{3,2,-1}[\tilde{\mu}, m_1] +$
$\frac{1}{8} m_1 \tilde{\mu} g_1^2 \left(3 y_d^{i1i2}(-2 g_1^2 + g_2^2) + 2 g_2^2 y_d^{pi2}\delta_{pi1}\right) LF_{4,1,-1}[\tilde{\mu}, m_1] -$
$\frac{1}{2} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} LF_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 \left(3 g_1^2 y_d^{i1i2} - 2 g_2^2 y_d^{pi2}\delta_{pi1}\right) LF_{5,1,-2}[\tilde{\mu}, m_1] +$
$\frac{1}{12} m_2 \tilde{\mu} g_2^2 \left(3 y_d^{i1i2}(3 g_1^2 + g_2^2) - 8 g_2^2 y_d^{pi2}\delta_{pi1}\right) LF_{3,1,0}[\tilde{\mu}, m_2] +$
$\frac{1}{3} m_2 \tilde{\mu} g_2^4 \left(9 y_d^{i1i2} - 2 y_d^{pi2}\delta_{pi1}\right) LF_{3,2,-1}[\tilde{\mu}, m_2] +$
$\frac{1}{8} m_2 \tilde{\mu} g_2^2 \left(-3 y_d^{i1i2}(6 g_1^2 + g_2^2) + 14 g_2^2 y_d^{pi2}\delta_{pi1}\right) LF_{4,1,-1}[\tilde{\mu}, m_2] -$
$2 m_2 \tilde{\mu} g_2^4 y_d^{i1i2} LF_{4,2,-2}[\tilde{\mu}, m_2] + \frac{1}{2} m_2 \tilde{\mu} g_2^2 \left(3 g_1^2 y_d^{i1i2} - 2 g_2^2 y_d^{pi2}\delta_{pi1}\right) LF_{5,1,-2}[\tilde{\mu}, m_2] +$
$\frac{1}{36} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,1,1,0}[m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] +$
$\frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,2,1,-1}[m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] - \frac{1}{36} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r}$
$LF_{3,1,1,-1}[m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,1,1,0}[m_1, m_{\tilde{d}}^{r}, \tilde{\mu}] -$
$\frac{1}{6} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{3,1,1,-1}[m_1, m_{\tilde{d}}^{r}, \tilde{\mu}] + \frac{1}{144} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 - g_2^2)$
$LF_{2,2,1,-1}[m_1, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}] - \frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{d}}^{r}] +$
$\frac{1}{144} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 - g_2^2) LF_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{d}}^{i2}] + \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}$
$LF_{2,1,1,0}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] - \frac{1}{36} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] -$
$\frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{3,1,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] + \frac{1}{36} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}$
$LF_{2,2,1,-1}[m_1, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{i1}] - \frac{1}{3} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,0}[m_1, m_{\tilde{u}}^{r}, \tilde{\mu}] +$
$\frac{1}{3} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{3,1,1,-1}[m_1, m_{\tilde{u}}^{r}, \tilde{\mu}] - \frac{5}{24} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r}$
$LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{d}}^{r}] + \frac{1}{24} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (-9 g_1^2 + g_2^2) LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{d}}^{i2}] +$
$\frac{1}{48} m_1 \tilde{\mu} g_1^2 \left(y_d^{i1i2}(-9 g_1^2 + g_2^2) - 2 y_d^{pi2}\left(\overline{y_d}^{pr} y_d^{i1r} + \overline{y_u}^{pr} y_u^{i1r}\right)\right) LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i1}] +$
$\frac{7}{24} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{u}}^{r}] + \frac{3}{4} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (3 m_1 + m_2)$
$LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2) LF_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] +$
$\frac{1}{8} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{d}}^{r}] - \frac{1}{2} m_2 \tilde{\mu} g_2^4 y_d^{pi2} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{q}}^{p}]\,\delta_{pi1} -$
$\frac{1}{16} m_2 \tilde{\mu} g_2^2 \left(y_d^{i1i2}(3 g_1^2 + 13 g_2^2) + 6 y_d^{pi2}\left(\overline{y_d}^{pr} y_d^{i1r} - \overline{y_u}^{pr} y_u^{i1r}\right)\right) LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}] -$
$\frac{1}{8} m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{u}}^{r}] - \frac{2}{3} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r}$
$LF_{2,1,1,0}[m_3, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] - \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,2,1,-1}[m_3, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] +$
$\frac{2}{3} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{3,1,1,-1}[m_3, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] + \frac{1}{6} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} (-g_1^2 + g_2^2)$
$LF_{2,2,1,-1}[m_3, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}] + \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{d}}^{r}] +$
$\frac{1}{6} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} (-g_1^2 + g_2^2) LF_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{d}}^{i2}] + \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}$
$LF_{2,1,1,0}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] - \frac{1}{3} m_3 \tilde{\mu} g_3^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] -$
$\frac{2}{3} m_3 \tilde{\mu} g_3^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{3,1,1,-1}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] + \frac{1}{3} m_3 \tilde{\mu} g_3^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}$
$LF_{2,2,1,-1}[m_3, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{i1}] + \frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} LF_{2,2,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}] -$
$\frac{1}{8} \tilde{\mu} y_d^{si2} y_d^{i1r} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} LF_{2,2,1,-1}[m_{\tilde{d}}^{r}, \tilde{\mu}, m_{\tilde{q}}^{p}] - \frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr}$
$LF_{2,2,1,-1}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] + \frac{1}{216} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 + 3 g_2^2) LF_{2,1,1,0}[m_{\tilde{d}}^{i2}, m_1, m_{\tilde{q}}^{i1}] +$
$\frac{1}{72} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 - g_2^2) LF_{3,1,1,-1}[m_{\tilde{d}}^{i2}, m_1, m_{\tilde{q}}^{i1}] + \frac{1}{18} m_1 \tilde{\mu} g_1^4 y_d^{i1i2}$
$LF_{2,1,1,0}[m_{\tilde{d}}^{i2}, m_1, \tilde{\mu}] - \frac{1}{9} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} (g_1^2 + 3 g_2^2) LF_{2,1,1,0}[m_{\tilde{d}}^{i2}, m_3, m_{\tilde{q}}^{i1}] +$
$\frac{1}{3} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} (-g_1^2 + g_2^2) LF_{3,1,1,-1}[m_{\tilde{d}}^{i2}, m_3, m_{\tilde{q}}^{i1}] + \frac{1}{4} \tilde{\mu} y_d^{i1i2} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr}$
$LF_{2,2,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}] - \frac{1}{4} \tilde{\mu} y_d^{i1i2} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} LF_{2,2,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] +$
$\frac{1}{2} \tilde{\mu} y_d^{i1i2} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} LF_{2,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}] - \frac{1}{2} \tilde{\mu} y_d^{i1i2} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr}$
$LF_{3,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}] + \frac{1}{4} \tilde{\mu} y_d^{i1i2} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} LF_{2,2,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}] +$
$\frac{3}{2} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} LF_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}] - \frac{3}{2} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr}$
$LF_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}] - 3 \tilde{\mu} y_d^{sr} y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} LF_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] +$
$3 \tilde{\mu} y_d^{sr} y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} LF_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] + \frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr}$
$LF_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}] - \frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} LF_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] +$
$\frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} LF_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] - \frac{3}{8} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr}$
$LF_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}] + \frac{1}{24} \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} (g_1^2 - 9 g_2^2) LF_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] +$
$\frac{1}{8} \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} (-g_1^2 + g_2^2) LF_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] +$
$3 \tilde{\mu} y_d^{sr} y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} LF_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}] +$
$\frac{1}{8} \tilde{\mu} y_d^{si2} y_d^{i1r} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} LF_{2,2,1,-1}[m_{\tilde{q}}^{p}, \tilde{\mu}, m_{\tilde{d}}^{r}] -$
$\frac{1}{16} \tilde{\mu} y_u^{i1r} \overline{a_u}^{pr} \left(y_d^{pi2}(g_1^2 - g_2^2) + 2 \overline{y_d}^{st} y_d^{pt} y_d^{si2}\right) LF_{2,2,1,-1}[m_{\tilde{q}}^{p}, \tilde{\mu}, m_{\tilde{u}}^{r}] -$
$\frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} LF_{2,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$
$\frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} LF_{3,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$
$\frac{9}{8} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} LF_{2,2,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] -$
$\frac{1}{216} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 + 3 g_2^2) LF_{2,1,1,0}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{d}}^{i2}] +$
$\frac{1}{72} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 - g_2^2) LF_{3,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{d}}^{i2}] +$
$\frac{1}{72} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 - 3 g_2^2) LF_{2,1,1,0}[m_{\tilde{q}}^{i1}, m_1, \tilde{\mu}] +$
$\frac{1}{8} m_2 \tilde{\mu} g_2^2 y_d^{i1i2} (g_1^2 + g_2^2) LF_{2,1,1,0}[m_{\tilde{q}}^{i1}, m_2, \tilde{\mu}] +$
$\frac{1}{9} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} (g_1^2 + 3 g_2^2) LF_{2,1,1,0}[m_{\tilde{q}}^{i1}, m_3, m_{\tilde{d}}^{i2}] +$
$\frac{1}{3} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} (-g_1^2 + g_2^2) LF_{3,1,1,-1}[m_{\tilde{q}}^{i1}, m_3, m_{\tilde{d}}^{i2}] +$
$\frac{3}{2} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} LF_{2,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] -$
$\frac{3}{2} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} LF_{3,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] +$
$\frac{1}{24} \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} (11 g_1^2 - 3 g_2^2) LF_{2,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] +$
$\frac{1}{8} \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} (-g_1^2 + g_2^2) LF_{3,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] +$
$\frac{1}{16} \tilde{\mu} y_u^{i1r} \overline{a_u}^{pr} \left(y_d^{pi2}(-g_1^2 + g_2^2) + 2 \overline{y_d}^{st} y_d^{pt} y_d^{si2}\right) LF_{2,2,1,-1}[m_{\tilde{u}}^{r}, \tilde{\mu}, m_{\tilde{q}}^{p}] -$
$\frac{3}{2} m_1 \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} LF_{2,1,1,0}[\tilde{\mu}, m_1, m_2] + \frac{1}{8} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (31 m_1 - 5 m_2) LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] +$
$\frac{3}{4} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (-3 m_1 + m_2) LF_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] + \frac{1}{2} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r}$
$LF_{2,1,1,0}[\tilde{\mu}, m_1, m_{\tilde{d}}^{r}] - \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{d}}^{r}] +$
$\frac{1}{12} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \left(\overline{y_d}^{pr} y_d^{i1r} + \overline{y_u}^{pr} y_u^{i1r}\right) LF_{2,1,1,0}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}] -$
$\frac{1}{12} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \left(\overline{y_d}^{pr} y_d^{i1r} + \overline{y_u}^{pr} y_u^{i1r}\right) LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}] -$
$\frac{1}{2} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,0}[\tilde{\mu}, m_1, m_{\tilde{u}}^{r}] +$
$\frac{1}{4} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{u}}^{r}] -$
$2 m_1 \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} LF_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] - \frac{1}{2} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,1,1,0}[\tilde{\mu}, m_2, m_{\tilde{d}}^{r}] +$
$\frac{1}{4} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{3,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{d}}^{r}] +$
$\frac{3}{4} m_2 \tilde{\mu} g_2^2 y_d^{pi2} \left(\overline{y_d}^{pr} y_d^{i1r} - \overline{y_u}^{pr} y_u^{i1r}\right) LF_{2,1,1,0}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}] +$
$\frac{3}{4} m_2 \tilde{\mu} g_2^2 y_d^{pi2} \left(-\overline{y_d}^{pr} y_d^{i1r} + \overline{y_u}^{pr} y_u^{i1r}\right) LF_{3,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}] +$
$\frac{1}{2} m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,0}[\tilde{\mu}, m_2, m_{\tilde{u}}^{r}] - \frac{1}{4} m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}$
$LF_{3,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{u}}^{r}] + \frac{1}{4} \tilde{\mu} y_d^{si2} y_d^{i1r} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} LF_{2,1,1,0}[\tilde{\mu}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] -$
$\frac{1}{4} \tilde{\mu} y_d^{si2} y_d^{i1r} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} LF_{3,1,1,-1}[\tilde{\mu}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] - \frac{1}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{si2} y_u^{i1r} \overline{a_u}^{pr}$
$LF_{2,1,1,0}[\tilde{\mu}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \frac{1}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{si2} y_u^{i1r} \overline{a_u}^{pr} LF_{3,1,1,-1}[\tilde{\mu}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$
$\frac{1}{3} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2) LF_{1,1,1,1,0}[m_1, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}] - \frac{1}{6} m_2 \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2}$
$LF_{2,1,1,1,-1}[m_1, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}] - \frac{1}{8} \tilde{\mu} g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] +$
$\frac{1}{16} \tilde{\mu} g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] - \frac{1}{36} \tilde{\mu} g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr}$
$LF_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}] + \frac{1}{72} \tilde{\mu} g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}] +$
$\frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, \tilde{\mu}] - \frac{1}{48} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r}$
$LF_{2,1,1,1,-1}[m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, \tilde{\mu}] - \frac{1}{9} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{i2}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] -$
$\frac{1}{3} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{i2}, m_{\tilde{d}}^{r}, \tilde{\mu}] + \frac{1}{9} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}$
$LF_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] - \frac{1}{9} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} LF_{2,1,1,1,-1}[m_1, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}, \tilde{\mu}] -$
$\frac{1}{3} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{i2}, m_{\tilde{u}}^{r}, \tilde{\mu}] - \frac{1}{8} \tilde{\mu} g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr}$
$LF_{1,1,1,1,0}[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] + \frac{1}{16} \tilde{\mu} g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] +$
$\frac{1}{36} \tilde{\mu} g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{1,1,1,1,0}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$
$\frac{1}{72} \tilde{\mu} g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$
$\frac{1}{8} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{1,1,1,1,0}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}, \tilde{\mu}] -$
$\frac{1}{48} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}, \tilde{\mu}] -$
$\frac{1}{6} m_1 \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} LF_{2,1,1,1,-1}[m_2, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}] +$
$\frac{3}{8} \tilde{\mu} g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{1,1,1,1,0}[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] -$
$\frac{3}{16} \tilde{\mu} g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] +$
$\frac{1}{4} \tilde{\mu} g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{1,1,1,1,0}[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}] -$
$\frac{1}{8} \tilde{\mu} g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}] -$
$\frac{3}{8} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{1,1,1,1,0}[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, \tilde{\mu}] +$
$\frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,1,1,1,-1}[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, \tilde{\mu}] -$
$\frac{3}{8} \tilde{\mu} g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{1,1,1,1,0}[m_2, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] +$
$\frac{3}{16} \tilde{\mu} g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_2, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] -$
$\frac{1}{4} \tilde{\mu} g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{1,1,1,1,0}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$
$\frac{1}{8} \tilde{\mu} g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$
$m_2 \tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{1,1,1,1,0}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, \tilde{\mu}] +$
$\frac{3}{8} m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{1,1,1,1,0}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}, \tilde{\mu}] -$
$\frac{1}{16} m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,1,-1}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}, \tilde{\mu}] -$
$\frac{4}{3} \tilde{\mu} g_3^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{1,1,1,1,0}[m_3, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}] +$
$\frac{2}{3} \tilde{\mu} g_3^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_3, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}] +$
$\frac{8}{3} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{1,1,1,1,0}[m_3, m_{\tilde{d}}^{i2}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] -$
$\frac{8}{3} m_3 \tilde{\mu} g_3^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{1,1,1,1,0}[m_3, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] +$
$\frac{4}{3} \tilde{\mu} g_3^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{1,1,1,1,0}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$
$\frac{2}{3} \tilde{\mu} g_3^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$
$\frac{1}{48} \tilde{\mu} g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{d}}^{r}, m_1, m_{\tilde{q}}^{p}, \tilde{\mu}] +$
$\frac{1}{48} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{d}}^{r}, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}] -$
$\frac{1}{16} \tilde{\mu} g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{d}}^{r}, m_2, m_{\tilde{q}}^{p}, \tilde{\mu}] -$
$\frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{d}}^{r}, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}] +$
$\tilde{\mu} y_d^{pi2} y_d^{sr} \overline{y_u}^{st} y_u^{i1t} \overline{a_d}^{pr} LF_{1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, \tilde{\mu}] -$
$\frac{3}{4} \tilde{\mu} y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} LF_{2,2,1,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] +$
$\frac{3}{4} \tilde{\mu} y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} LF_{2,2,1,1,-1}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] -$
$\frac{1}{4} \tilde{\mu} y_d^{i1i2} y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} LF_{2,2,1,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] +$
$\frac{1}{4} \tilde{\mu} y_d^{i1i2} y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} LF_{2,2,1,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] +$
$\frac{1}{2} \tilde{\mu} y_d^{i1i2} y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} LF_{2,1,1,1,0}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] -$
$\frac{1}{2} \tilde{\mu} y_d^{i1i2} y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} LF_{3,1,1,1,-1}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] -$
$\frac{1}{72} \tilde{\mu} g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{p}, m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] -$
$\frac{1}{48} \tilde{\mu} g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{p}, m_1, m_{\tilde{d}}^{r}, \tilde{\mu}] +$
$\frac{1}{72} \tilde{\mu} g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{p}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] -$
$\frac{1}{48} \tilde{\mu} g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{p}, m_1, m_{\tilde{u}}^{r}, \tilde{\mu}] +$
$\frac{1}{8} \tilde{\mu} g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{p}, m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] +$
$\frac{1}{16} \tilde{\mu} g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{p}, m_2, m_{\tilde{d}}^{r}, \tilde{\mu}] -$
$\frac{1}{8} \tilde{\mu} g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{p}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] -$
$\frac{1}{16} \tilde{\mu} g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{p}, m_2, m_{\tilde{u}}^{r}, \tilde{\mu}] -$
$\frac{2}{3} \tilde{\mu} g_3^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{p}, m_3, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] +$
$\frac{2}{3} \tilde{\mu} g_3^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{p}, m_3, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] +$
$3 \tilde{\mu} y_d^{i1i2} y_u^{pt} \overline{a_d}^{pr} \left(-a_d^{sr} \overline{a_u}^{st} + \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}\right) LF_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +$
$3 \tilde{\mu} y_d^{i1i2} y_u^{pt} \overline{a_d}^{pr} \left(a_d^{sr} \overline{a_u}^{st} - \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}\right) LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +$
$\frac{3}{4} \tilde{\mu} y_d^{i1i2} y_u^{pt} \overline{a_d}^{pr} \left(a_d^{sr} \overline{a_u}^{st} - 3 \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}\right) LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] +$
$\frac{3}{4} \tilde{\mu}^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} LF_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] -$
$\frac{3}{4} \tilde{\mu}^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] +$
$\tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{si2} y_u^{i1r} \overline{a_u}^{pr} LF_{1,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, \tilde{\mu}] -$
$\frac{3}{8} \tilde{\mu}^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +$
$3 \tilde{\mu} y_d^{i1i2} y_u^{pt} \overline{a_d}^{pr} \left(a_d^{sr} \overline{a_u}^{st} - \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}\right) LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] +$
$\frac{3}{8} \tilde{\mu}^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] +$
$\frac{3}{2} \tilde{\mu} y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} LF_{2,1,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] -$
$\frac{3}{2} \tilde{\mu} y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} LF_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] +$
$\frac{3}{4} \tilde{\mu}^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} LF_{2,1,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] -$
$\frac{3}{4} \tilde{\mu}^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} LF_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] -$
$\frac{3}{8} \tilde{\mu}^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}] -$
$\frac{9}{8} \tilde{\mu}^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$
$\frac{1}{72} \tilde{\mu} g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] -$
$\frac{1}{48} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{d}}^{r}, \tilde{\mu}] -$
$\frac{1}{72} \tilde{\mu} g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$
$\frac{1}{48} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{u}}^{r}, \tilde{\mu}] -$
$\frac{1}{8} \tilde{\mu} g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] +$
$\frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{d}}^{r}, \tilde{\mu}] +$
$\frac{1}{8} \tilde{\mu} g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$
$\frac{1}{16} m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{u}}^{r}, \tilde{\mu}] +$
$\frac{2}{3} \tilde{\mu} g_3^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_3, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] -$
$\frac{2}{3} \tilde{\mu} g_3^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_3, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$
$\frac{1}{48} \tilde{\mu} g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{u}}^{r}, m_1, m_{\tilde{q}}^{p}, \tilde{\mu}] +$
$\frac{1}{48} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{u}}^{r}, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}] +$
$\frac{1}{16} \tilde{\mu} g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[m_{\tilde{u}}^{r}, m_2, m_{\tilde{q}}^{p}, \tilde{\mu}] +$
$\frac{1}{16} m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{u}}^{r}, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}] -$
$\frac{3}{4} \tilde{\mu}^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} LF_{2,2,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] +$
$\frac{3}{2} \tilde{\mu}^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} LF_{2,1,1,1,0}[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] -$
$\frac{3}{2} \tilde{\mu}^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} LF_{3,1,1,1,-1}[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] +$
$\frac{1}{6} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (-3 m_1 + m_2) LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_{\tilde{d}}^{i2}] -$
$\frac{1}{12} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (m_1 + 9 m_2) LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_{\tilde{q}}^{i1}] +$
$\frac{25}{48} \tilde{\mu} g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] -$
$\frac{1}{16} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] +$
$\frac{25}{48} \tilde{\mu} g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$
$\frac{1}{2} m_1 g_1^2 \tilde{\mu}^3 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,1,0}[\tilde{\mu}, m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$
$\frac{1}{16} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] +$
$\frac{7}{16} \tilde{\mu} g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] +$
$\frac{3}{16} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}] +$
$\frac{9}{16} \tilde{\mu} g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$
$\frac{1}{2} m_2 g_2^2 \tilde{\mu}^3 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,1,0}[\tilde{\mu}, m_2, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$
$\frac{3}{16} m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}] +$
$\frac{1}{6} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2) LF_{1,1,1,1,1,-1}[m_1, m_2, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}, \tilde{\mu}] +$
$\frac{2}{3} m_1 m_2 \tilde{\mu} g_1^2 g_2^2 a_d^{i1i2} LF_{1,1,1,1,1,0}[m_1, m_2, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}, \tilde{\mu}] +$
$\frac{1}{6} \tilde{\mu} g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{1,1,1,1,1,-1}[m_1, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, \tilde{\mu}] -$
$\frac{1}{3} m_1 \tilde{\mu} g_1^2 y_d^{i1r} \overline{a_d}^{pr} a_d^{pi2} LF_{1,1,1,1,1,0}[m_1, m_{\tilde{d}}^{i2}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] -$
$\frac{1}{9} m_1 \tilde{\mu} g_1^2 y_d^{i1r} \overline{a_d}^{pr} a_d^{pi2} LF_{1,1,1,1,1,0}[m_1, m_{\tilde{d}}^{i2}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}] +$
$\frac{1}{3} m_1 \tilde{\mu} g_1^2 y_u^{i1r} \left(-a_d^{pi2} \overline{a_u}^{pr} + \tilde{\mu}^2 y_d^{pi2} \overline{y_u}^{pr}\right) LF_{1,1,1,1,1,0}[m_1, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] +$
$\frac{1}{9} m_1 \tilde{\mu} g_1^2 y_u^{i1r} \left(a_d^{pi2} \overline{a_u}^{pr} - \tilde{\mu}^2 y_d^{pi2} \overline{y_u}^{pr}\right) LF_{1,1,1,1,1,0}[m_1, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$
$\frac{1}{6} \tilde{\mu} g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{1,1,1,1,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] -$
$\frac{1}{6} m_1 g_1^2 \tilde{\mu}^3 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{1,1,1,1,1,0}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] -$
$\frac{1}{2} \tilde{\mu} g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} LF_{1,1,1,1,1,-1}[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, \tilde{\mu}] -$
$m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{a_d}^{pr} a_d^{i1r} LF_{1,1,1,1,1,0}[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, \tilde{\mu}] +$
$\frac{1}{2} \tilde{\mu} g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{1,1,1,1,1,-1}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] -$
$\frac{1}{2} m_2 g_2^2 \tilde{\mu}^3 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{1,1,1,1,1,0}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] +$
$\frac{8}{3} m_3 \tilde{\mu} g_3^2 y_d^{i1r} \overline{a_d}^{pr} a_d^{pi2} LF_{1,1,1,1,1,0}[m_3, m_{\tilde{d}}^{i2}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}] +$
$\frac{8}{3} m_3 \tilde{\mu} g_3^2 y_u^{i1r} \left(-a_d^{pi2} \overline{a_u}^{pr} + \tilde{\mu}^2 y_d^{pi2} \overline{y_u}^{pr}\right) LF_{1,1,1,1,1,0}[m_3, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$
$\tilde{\mu} y_d^{pi2} y_u^{i1t} \overline{a_d}^{pr} \left(a_d^{sr} \overline{a_u}^{st} - \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}\right) LF_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, \tilde{\mu}] +$
$\tilde{\mu}^3 y_d^{pi2} \overline{y_u}^{st} y_u^{sr} y_u^{i1t} \overline{a_u}^{pr} LF_{1,1,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, \tilde{\mu}]\Big)$